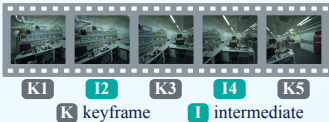


Online observations

RGB stream

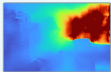


Online
frontend

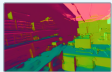
Estimated poses



Keyframe depth



Keyframe normal



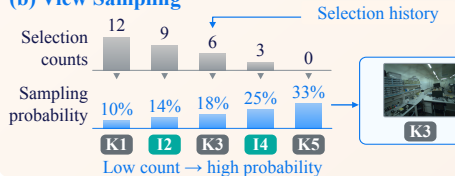
Training-view management

(a) View Set Growth

Retain old views; admit arrived views



(b) View Sampling



Gaussian map optimization

Shared Gaussian map

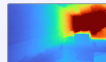
Gaussian
initialization



Update

Camera pose

Render



Rendered RGB

Rendered depth

Rendered normal

Map
objective
+

Priors
RGB

Base supervision

RGB (K + I) · Depth / normal (K)

(c) Free-Space Carve Loss

